Practice Set 2 Paper 3 – Subspecialities of Urology

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Practice paper compiled from the question bank.

Q1. Iatrogenic ureteric injury

Type: LAQ (25 marks) (upper ureteric injury also answered in 2023-Winter Paper 3, Q2)

Assumed scope: surgical (iatrogenic) injury at any level of the ureter, in gynaecological, colorectal, vascular and urological surgery.

Introduction

- Iatrogenic injury is commoner than external trauma. **The distal ureter is the commonest site** in non-urological surgery.
- It is mainly a problem of **recognition**: thermal, ligation and devascularisation injuries are invisible at operation and declare themselves days to weeks later.

Aetiology and incidence (EAU (European Association of Urology) trauma data)

Source	Key facts
Gynaecological	Historically about half. Hysterectomy: vaginal 0.02–0.5%, abdominal 0.03–2%, minimally invasive 0.13–6%; prolapse surgery highest (1.7–4.3%). Endometriosis and placenta accreta raise risk
Colorectal	0.15–10%; laparoscopic equals open
Urological	Ureteroscopy is the commonest cause in some series: perforation 0.2–2%, avulsion under 0.5% (proximal third most at risk); stricture about 3%. Also radical prostatectomy, partial nephrectomy or ablation (late proximal stricture)
Vascular	Aorto-iliac bypass: transient hydronephrosis 12–20%; up to 85% unrecognised. Rare ureteroarterial fistula (chronic stent plus pelvic surgery or radiation)

- **Danger sites in the pelvis:** pelvic brim near the infundibulopelvic ligament; under the uterine artery; beside the uterosacral ligament; vaginal fornix near the UVJ (ureterovesical junction).
- **Mechanisms:** ligation, crush, laceration, transection, avulsion, thermal injury, devascularisation. About 10% of women get a large share of distal ureteric blood from the uterine artery, so some injuries are unavoidable.
- **Risk factors:** previous surgery, adhesions, malignancy, radiotherapy, heavy bleeding, long operation.
- **URS (ureteroscopy) risks:** long operation, 12/14-Fr sheath, impacted or proximal stone, inexperience.

Classification: AAST (American Association for the Surgery of Trauma)

Grade	Injury
I	Haematoma or contusion, no devascularisation
II	Laceration <50% of circumference
III	Laceration ≥50%
IV	Complete transection, <2 cm devascularised
V	Avulsion, >2 cm devascularised

Prevention

- See the ureter at the **pelvic brim** (crossing the common iliac artery) and trace it before ligating pedicles. Good haemostasis and exposure.
- **Prophylactic stents** help recognition in high-risk cases, but do not clearly reduce injury. **ICG (indocyanine green)** instilled into the ureter makes it fluoresce for hours.

- URS: safety wire; ≤14-Fr sheath in an unstented ureter; fragment before basketing; "stop, stent, stage".

Recognition

Intraoperative (best outcome)

- Inspect the ureter; IV (intravenous) indigo carmine or methylene blue; cystoscopy for efflux, with retrograde pyelogram or stent. Check the other ureter and bladder.
- Methylene blue: avoid with SSRIs (selective serotonin reuptake inhibitors; serotonin toxicity) and G6PD (glucose-6-phosphate dehydrogenase) deficiency.

Delayed (most cases)

- Fever, ileus, flank pain, leucocytosis, rising drain output, urinoma, anuria (bilateral), or **ureterovaginal fistula**: normal voiding plus constant leak, usually within 4 weeks.
- **CT urography with 10-minute delays** (both ureters). **Retrograde pyelogram** is most sensitive and allows a stent. Drain-fluid creatinine above serum confirms urine. Double dye test separates VVF (vesicovaginal fistula) from ureterovaginal fistula.

Management

Principles: debride to bleeding tissue; spatulated, tension-free, watertight, stented repair with fine absorbable suture; drain; preserve adventitia; omentum for thermal or radiation injury.

By timing

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Recognised at surgery -> repair at once (stable patient)
Unstable -> ligate + later nephrostomy, or exteriorised stent
Recognised late -> try retrograde stent (or nephrostomy + antegrade stent)
   |-- succeeds -> stent 4-8 weeks, then image
   |-- fails -> nephrostomy; early repair if <=7 days,
               otherwise delayed repair after about 6 weeks
Uncontrolled sepsis / collections -> drain; repair regardless of timing
```

At surgery, by injury

- Suture through the wall or minor thermal injury: remove suture, stent.
- Partial tear: close transversely over a stent (4–8 weeks).
- Ligation: remove ligature; stent if viable, excise and repair if not.

By level (complete injury)

Level	Repair
Upper	Ureteroureterostomy; ureteropyelostomy (PUJ (pelviureteric junction) avulsion); ureterocalicostomy
Mid	Ureteroureterostomy ± downward nephropexy; TUU (transureteroureterostomy) when delayed
Lower (commonest iatrogenic)	**Ureteroneocystostomy, non-tunnelled ± psoas hitch** (95–100% success); **Boari flap** for gaps up to 12 cm
Long loss (delayed only)	Ileal ureter or Yang–Monti; buccal graft ureteroplasty; autotransplantation (last resort)

- **URS perforation:** stop, stent about 4 weeks, image later. **Avulsion:** nephrostomy, delayed reconstruction; avoid nephrectomy.
- **Ureterovaginal fistula:** stent first (best within 2 weeks); reimplant if it fails (success ≥90%).
- **Ureteroarterial fistula:** endovascular covered stent.

Follow-up

- Ultrasound or diuretic renogram at 4–12 weeks and 1 year; strictures can be silent.
- **Also know:** robotic reconstruction for delayed repair; malpractice claims follow gynaecological surgery most often.

Diagram to draw: psoas hitch with non-tunnelled reimplant; ureter's relation to the uterine artery.

> **Key points to remember:**
>
> - Distal ureter commonest iatrogenic site; most injuries present late.
> - CT with 10-minute delays; retrograde pyelogram is most sensitive and allows stenting.
> - Late: stent first; early repair ≤7 days, otherwise delayed after 6 weeks.
> - Lower ureter: psoas hitch 95–100%; Boari flap up to 12 cm.

Q2. Discuss mgt. of b/l VUR.

Type: LAQ (25 marks) (also answered in 2015-Summer Paper 3, Q2 and 2020-Summer Paper 1, Q2)

Introduction

- **VUR (vesicoureteral reflux):** retrograde flow of urine from bladder to upper tract. **Primary:** short intravesical tunnel; tunnel length to diameter should be **5:1** (Paquin). **Secondary:** high bladder pressure (PUV (posterior urethral valves), neurogenic bladder, BBD (bladder and bowel dysfunction)).
- **Bilateral VUR is a high-risk feature:** it resolves less and scars more. **Infection, not sterile reflux, damages the kidney.** So management treats infection risk, not the reflux itself.

International grading (VCUG, voiding cystourethrogram)

Grade	Findings
I	Non-dilated ureter only
II	Pelvis and calyces reached, no dilatation
III	Mild–moderate dilatation, minimal forniceal blunting
IV	Moderate dilatation and tortuosity
V	Gross dilatation; papillary impressions lost

Evaluation

- Febrile UTI (urinary tract infection) history, BBD, BP (blood pressure), growth, creatinine, proteinuria.
- **RBUS (renal and bladder ultrasound)**; **VCUG** (grade, urethra: excludes PUV); **DMSA (dimercaptosuccinic acid)** for scars and split function. Nuclear cystogram for follow-up.
- **UDR (ureteral diameter ratio)** predicts resolution and breakthrough UTI better than grade.

Natural history

- Grade II 63–85% resolve; grade III about 50%; **grades IV–V and bilateral III rarely**.
- **Bilateral grade IV persists most** (AUA (American Urological Association) 1997). Infants with high grade still resolve fairly well.

Risk stratification (Hidas)

Low risk	High risk
Male, >5 years	Female, <1 year
Unilateral, grade I–II	**Bilateral**, grade III–V
No BBD, no scars	BBD, scars, multiple febrile UTIs

Medical management

- **Treat BBD first:** timed voiding, hydration, constipation treatment, anticholinergics if needed.
- **CAP (continuous antibiotic prophylaxis):** one low dose at night.

Agent	Age	Note
Trimethoprim or amoxicillin	<2 months	Avoid sulfonamide (jaundice)

Agent	Age	Note
TMP-SMX (trimethoprim-sulfamethoxazole)	>2 months	Used in RIVUR
Nitrofurantoin	>2 months	Rare lung, liver toxicity

- **RIVUR (2014):** recurrent UTI **23.6% → 12.9%**; **scarring unchanged**; benefit mainly in high risk.
- **Swedish Reflux Trial** (grade III–IV, 1–2 years): girls had febrile UTI 19% on CAP, 23% after endoscopy, 57% on surveillance; boys no difference.
- **IRSC (International Reflux Study in Children) and Birmingham:** surgery equalled medical care for new scars.
- **Circumcision** or topical steroid for phimosis (NNT (number needed to treat) about 4).
- **Guidelines:** AUA 2010 and EAU/ESPU (European Association of Urology/European Society for Paediatric Urology): CAP for high risk; correct after breakthrough infection.

Indications for correction

- Breakthrough febrile UTI despite good bladder and bowel habits.
- New scars or falling function.
- Persistent high-grade (IV–V) bilateral VUR (girls traditionally corrected before puberty; benefit unproven).
- Non-adherence or intolerance to CAP; parental choice.
- **Controversy:** asymptomatic bilateral high-grade VUR in an older boy with normal kidneys can be observed.

Surgical techniques

Principles: exclude secondary VUR; gentle handling; tension-free; tunnel 5:1; straight course.

1. Endoscopic injection (Dx/HA, dextranomer/hyaluronic acid)

- **STING** (subureteric Teflon injection): 2–3 mm below the orifice. **HIT** (hydrodistension implantation): inside the orifice, better. **Double HIT** about 90%.
- Aim for a "volcano" mound; day case. Repeat once or twice, then reimplant.
- Recurrence about 20% at 2 years; late obstruction reported.

2. Open reimplantation (success **95.9%**, Elder 1997)

Technique	Method	Strength / weakness
Cohen	Intravesical, cross-trigonal	Commonest, over 98%; later ureteroscopy hard
Politano–Leadbetter	Intravesical, new superomedial hiatus	Long tunnel; blind hiatus risks bowel
Glenn–Anderson	Intravesical, advanced to bladder neck	Least kinking; short tunnel
Lich–Gregoir	Extravesical detrusorotomy	No haematuria or spasm; **retention up to 10% after bilateral**
RALUR-EV (robot-assisted extravesical)	Y-shaped detrusorotomy	Less pain; approaching open results

Criteria for selection

Situation	Preferred option
Grades II–IV, parents want least invasive	Dx/HA injection (lower success at higher grade)
Bilateral VUR, small or trabeculated bladder	**Cohen** (both sides through one cystotomy)

Situation	Preferred option
Grade V, dilated ureter	Intravesical reimplant ± tapering (Politano–Leadbetter or Cohen)
Unilateral, older child	Lich–Gregoir or RALUR-EV
Bilateral extravesical	Nerve-sparing dissection; warn about retention
Duplex system	Common sheath reimplant
Failed injection	Reimplant (not harder after Dx/HA)
Neurogenic or PUV bladder	Treat bladder first (anticholinergics, CIC (clean intermittent catheterisation))

Complications and follow-up

- Ultrasound at 6–12 weeks. Persistent reflux usually resolves by 1 year. Obstruction early (oedema): stent or nephrostomy; late (ischaemia, kink): redo.
- Lifelong BP checks; scarred women risk pre-eclampsia.

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Treat BBD -> low risk: observe | high risk (bilateral): CAP
Breakthrough UTI / new scar / persistent IV-V -> Dx/HA or Cohen
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Diagram to draw: grades I–V; Cohen cross-trigonal reimplant.

> **Key points to remember:**
>
> - Bilateral grade IV persists most; bilateral VUR is high risk.
> - RIVUR: CAP halves recurrent UTI, not scarring.
> - Open reimplant 95.9%; Cohen over 98% and suits bilateral VUR.
> - Bilateral extravesical repair: retention up to 10%.

Q3a. Uroflowmetry

Type: SAQ (10 marks) (also answered in 2018-Summer Paper 2, Q3f)

Definition

- Non-invasive urodynamic test recording **voided volume per unit time** (mL/s). It shows detrusor contraction and outlet resistance together.

Method

- Comfortably full bladder; private void in the preferred position. Ask whether the void was typical; compare with the frequency–volume chart.
- **Voided volume ≥150 mL** (Qmax unreliable below 125–150 mL). Repeat; then measure **PVR (postvoid residual)** by ultrasound.
- Flowmeters: weight transducer (load cell), rotating disc, capacitance dipstick.

Parameters

Parameter	Note
Qmax (maximum flow)	Key value; read from smoothed curve, ignoring spikes
Qave (average flow)	Less accurate for obstruction
Voided volume	≥150 mL
Flow time vs voiding time	Voiding time includes interruptions
Time to Qmax	Normally early in the void
Voiding efficiency	VV ÷ (VV + PVR) × 100

Curve patterns (ICCS (International Children's Continence Society))

Pattern	Suggests
Bell	Normal
Tower (high, short)	Overactive bladder
Plateau (flat, prolonged)	Fixed obstruction: stricture, PUV (posterior urethral valves)
Staccato (never reaches zero)	Sphincter overactivity, dysfunctional voiding
Interrupted (reaches zero)	Detrusor underactivity with straining, or DSD (detrusor-sphincter dyssynergia)

Interpretation

- **Qmax <10 mL/s:** obstruction likely (more specific). **<15 mL/s:** more sensitive, less specific.
- **Volume correction:** corrected Qmax = Qmax ÷ √VV; **Siroky** (men) and **Liverpool** (men and women) nomograms.
- **Key limitation:** cannot separate BOO (bladder outlet obstruction) from DUA (detrusor underactivity); only a pressure–flow study can.
- Children: valid if voided volume ≥50% of expected capacity; Qmax only if the peak lasts ≥2 s.

Uses

- BPH (benign prostatic hyperplasia): before active treatment and to monitor outcome.
- After urethroplasty or urethrotomy (stricture recurrence).

- Children: BBD (bladder and bowel dysfunction), after valve ablation; with patch EMG (electromyography).
- Women with voiding dysfunction.
- **AUA (American Urological Association):** pressure–flow study may be done before BPH surgery if Qmax >10 mL/s.

Diagram to draw: bell, plateau, tower and interrupted curves.

> **Key points to remember:**
> - Voided volume ≥150 mL; Qmax is the key value.
> - Plateau = fixed obstruction; staccato = sphincter; interrupted = weak detrusor.
> - Cannot distinguish BOO from DUA.

Q3b. Ejacultory duct obstruction

Type: SAQ (10 marks)

Assumed meaning: ejaculatory duct obstruction (EDO).

Anatomy

- Intraprostatic continuations of the seminal vesicles, joined by the vasal ampullae; they lack muscle and open at the **verumontanum**. Seminal vesicles give about 60% of the ejaculate and its fructose.

Aetiology

- **Congenital (commonest):** midline (Müllerian/utricular) cyst, Wolffian duct cyst, atresia. Idiopathic EDO carries CFTR (cystic fibrosis transmembrane conductance regulator) mutations in about half.
- **Acquired:** infection, instrumentation, duct stones, compression.
- **Zinner syndrome:** seminal vesicle cyst, ipsilateral renal agenesis, EDO.
- Complete bilateral EDO causes under 5% of azoospermia. Partial EDO gives oligoasthenospermia.

Clinical features

- Infertility, painful ejaculation, haematospermia, perineal pain.

Diagnosis

Semen (complete bilateral EDO)

- Volume **<1.0 mL**, **acidic pH**, low or absent **fructose**, azoospermia.
- Palpable vas, normal FSH (follicle-stimulating hormone) and testes.
- Exclude retrograde ejaculation and CBAVD (congenital bilateral absence of the vas deferens).

Investigations

Test	Method / finding
TRUS (transrectal ultrasound)	Seminal vesicle AP diameter **>1.5 cm**, duct >2.3 mm, midline cyst. Low specificity
Seminal vesicle aspiration	**≥3 sperm per high-power field** confirms obstruction
Vesiculography	Contrast into vesicles; invasive
Chromotubation	Indigo carmine into vesicles; dye at the veru on cystoscopy

Management

TURED (transurethral resection of the ejaculatory ducts)

- For azoospermic or severely oligoasthenospermic men, after the couple declines IVF (in vitro fertilisation).
- **Steps:** TRUS-guided aspiration (cryopreserve sperm) → instil indigo carmine with contrast → 24-Fr resectoscope, finger in rectum → resect the veru until dye effluxes. Spare bladder neck, sphincter and rectum. Catheter overnight.
- **Results:** volume rises in most; sperm appear in about half. Cold-knife incision alone re-obstructs.

Complications

- **Urine reflux** into the ducts, **epididymitis**, retrograde ejaculation, rectal injury, re-obstruction.

Alternatives

- **Seminal vesiculoscopy** (4–6 Fr via utricle): avoids reflux; restenosis unknown. Sperm retrieval with ICSI (intracytoplasmic sperm injection).

Diagram to draw: sagittal prostate with midline cyst compressing the ducts at the veru.

> **Key points to remember:**
>
> - Low volume (<1 mL), acidic, fructose-negative azoospermia with palpable vasa.
> - TRUS: vesicle >1.5 cm; aspiration ≥3 sperm per high-power field confirms.
> - TURED: sperm return in about half; reflux and epididymitis are its price.

Q3c. Ureterocele

Type: SAQ (10 marks) (ectopic ureter also answered in 2018-Summer Paper 3, Q1)

Definition

- Cystic dilatation of the distal ureter, in the bladder or spanning bladder neck and urethra; **never wholly urethral**, never joined to Wolffian structures.
- Commoner in girls (about 4:1); about 80% drain the **upper pole of a duplex kidney** (Weigert–Meyer rule: upper pole orifice caudal-medial); about 10% bilateral.
- **Cause:** persistent **Chwalla membrane**; upper pole often dysplastic.

Classification

System	Basis	Types
Glassberg (most useful)	Position	**Intravesical** vs **ectopic (extravesical)**: part at bladder neck or urethra
Stephens	Orifice morphology	Stenotic, sphincteric, sphincterostenotic, **cecoureterocele** (cavity extends under urethra), blind, non-obstructed
Churchill	Function	Upper pole only / whole kidney / contralateral at risk

Clinical features

- Antenatal hydronephrosis (most); febrile UTI (urinary tract infection) or urosepsis; adult single-system type with stone.
- **Prolapse:** smooth interlabial mass in girls; BOO (bladder outlet obstruction).
- **No incontinence when untreated** (unlike female ectopic ureter).

Evaluation

- **Ultrasound:** thin-walled cyst in bladder, dilated upper pole and ureter.
- **VCUG (voiding cystourethrogram):** lower pole reflux ("drooping lily"), eversion (weak floor), prolapse.
- **DMSA (dimercaptosuccinic acid):** upper pole function; cystoscopy before incision.

Management

- **Goals:** preserve function; remove infection, obstruction, reflux; keep continence.
- **CAP (continuous antibiotic prophylaxis)** from birth if hydronephrotic.
- **Observation:** small, non-obstructing, low-grade reflux, non-functioning upper pole.

Option	Indication / result
TUI (transurethral incision)	Sepsis, neonates, intravesical type. Low transverse full-thickness incision; relieves obstruction in 78–97%; new reflux 0–50%. Intravesical without reflux: 70–80% cured; ectopic: secondary surgery up to 70%
Upper pole heminephrectomy	Non-functioning upper pole, no reflux; secondary surgery 40–50%
Ureteroureterostomy / pyeloureterostomy	Functioning upper pole
Excision + common sheath reimplant ± bladder neck repair	Persistent reflux, ectopic or cecoureterocele; persisting reflux 5–10%
Total reconstruction	Older child, non-functioning upper pole, lower pole reflux

- **Also know:** "watering-can" laser punctures (less reflux); pseudoureterocele; ureterocele disproportion.

Diagram to draw: duplex kidney with upper pole ureterocele and lower pole reflux.

Key points to remember:

- Upper pole of duplex; never wholly urethral; Chwalla membrane.
- Intravesical vs ectopic predicts TUI success.
- TUI relieves obstruction in 78–97%; new reflux 0–50%.

Q3d. Discuss treatment of b/l VUR. Describe surgical techniques and criteria for their selection

Type: SAQ (10 marks)

See Practice Set 2 Paper 3, Q2 (full answer).

Key points to remember (SAQ angle):

- Treat BBD (bladder and bowel dysfunction) and give CAP (continuous antibiotic prophylaxis) first; correct for breakthrough febrile UTI (urinary tract infection), new scars or persistent grade IV–V.
- Endoscopic Dx/HA (dextranomer/hyaluronic acid) by HIT (hydrodistension implantation): least invasive; lower success at higher grades.
- Bilateral VUR (vesicoureteral reflux): Cohen cross-trigonal reimplant (over 98%) does both sides through one cystotomy; bilateral extravesical (Lich–Gregoir) repair risks retention up to 10%.

Q3e. Darifenacin vs. solifenacin vs. tolterodine

Type: SAQ (10 marks) (also answered in Practice Set 1 Paper 1, Q3e)

Introduction

- Antimuscarinics treat OAB (overactive bladder) after behavioural therapy. Acetylcholine on **M3** receptors contracts the detrusor; M2 (commoner) adds to it. Blockade reduces involuntary contractions and afferent urgency signals.
- Side effects: **M1** (brain) cognition; **M2** heart rate; **M3** (glands, gut) dry mouth, constipation.
- Selectivity is by **receptor subtype** (M3) or **tissue** (bladder over salivary gland); tissue selectivity matters more.

Comparison

Feature	Darifenacin	Solifenacin	Tolterodine
Selectivity	**Most M3-selective (about 30-fold)**; not tissue-selective	M3 about 10-fold over M2; bladder-selective	Non-subtype-selective; **bladder-selective**
Dose	7.5–15 mg daily (ER (extended-release))	5–10 mg daily	2 mg twice daily (IR (immediate-release)); 4 mg daily (ER)
Half-life	Moderate (ER once daily)	**Long** (about 2–3 days): once daily	Short; active metabolite 5-HMT (5-hydroxymethyl tolterodine)
Metabolism	CYP3A4 (a liver enzyme) and CYP2D6	CYP3A4	CYP2D6 (and 3A4)
Dry mouth	Dose-related	Dose-related	Lower with ER
Constipation	**Highest** (gut M3)	Moderate–high	Lower
CNS (central nervous system)	Spares M1: **preferred in elderly**	Better persistence	High dose: QT prolongation
Dose limit	Lower dose with CYP3A4 inhibitors or liver disease	5 mg in severe renal or moderate liver impairment	Halve with renal or liver impairment

Common points

- **Efficacy is similar**; response is individual, so switch agent or dose.
- **STAR trial:** flexible solifenacin beat tolterodine ER for urgency and incontinence.
- **Contraindications:** urinary or gastric retention, uncontrolled narrow-angle glaucoma.
- Men: add to α-blocker; retention up to 3%; check PVR (postvoid residual).
- **Dementia:** use >3 months raises risk 10–65% (SUFU (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction)). Prefer **β3 agonist**, or ER trospium, darifenacin or fesoterodine, in the elderly. Beers criteria: avoid in dementia.
- **Combination:** solifenacin plus mirabegron beats monotherapy (SYNERGY).

Diagram to draw: muscarinic subtypes M1 (brain), M2 (heart), M3 (bladder, gland, gut) with each drug's selectivity.

Key points to remember:

- Darifenacin most M3-selective; tolterodine bladder-selective; solifenacin longest half-life.
- Similar efficacy; STAR: solifenacin beat tolterodine ER.
- Elderly: darifenacin, trospium or β3 agonist; dementia risk 10–65% with >3 months' use.

Q3f. Primary obst. megaureter.

Type: SAQ (10 marks)

Assumed meaning: primary obstructive megaureter (POM).

Definition

- **Megaureter:** ureter diameter **≥7 mm** (normal up to 16 years is 5.0–6.5 mm). It is a descriptive term.
- **Classification:** primary (intrinsic) or secondary (neurogenic bladder, outlet obstruction); and **obstructed, refluxing, non-refluxing non-obstructed, or refluxing and obstructed**.
- **POM:** an **adynamic juxtavesical segment** at the UVJ (ureterovesical junction) blocks peristalsis and causes functional obstruction above.

Pathogenesis

- Ureteric muscle matures cranially to caudally; the juxtavesical segment matures last, even after birth. This explains spontaneous resolution.
- Excess collagen, thick muscle sleeve, abnormal interstitial cells of Cajal.

Clinical features

- 10–23% of antenatal upper-tract dilatation. Boys, **left side**; **bilateral 25%**; contralateral dysplasia or obstruction 10–15%.
- Mostly asymptomatic; or UTI (urinary tract infection), pain, haematuria, stones.

Evaluation

- **Ultrasound:** hydroureteronephrosis.
- **VCUG (voiding cystourethrogram):** excludes reflux and PUV (posterior urethral valves).
- **MAG3 (mercaptoacetyltriglycine) renogram:** split function; washout T½ unreliable (false positive up to 44%).
- MRU (magnetic resonance urography) for anatomy.

Management

Conservative (most)

- CAP (continuous antibiotic prophylaxis) in infants; serial ultrasound and renograms; most resolve.
- Diameter <10 mm: virtually never needs surgery; ≥10 mm: a substantial minority do.
- Follow to early adulthood.

Indications for surgery

- Recurrent febrile UTI; progressive dilatation; **DRF (differential renal function) <40%**; **fall in DRF ≥5%**.

Infant (small bladder, huge ureter): temporise

- Cutaneous ureterostomy; refluxing reimplant; nephrostomy; **endoscopic balloon dilatation with double-J** for 2–6 months (70–80% success; complications up to a third).

Definitive repair (child)

- Excise stenotic segment → straighten → **taper** → **non-refluxing reimplant at 5:1** (Cohen or Politano–Leadbetter; extravesical for huge ureters). Stent afterwards.

Tailoring	Method	Use
Starr plication	Imbricate lateral wall	Moderate (<1.75 cm)

Tailoring	Method	Use
Kalicinski folding	Exclude and fold lumen	Moderate
Hendren excision	Excise lateral wall	Massive; risk of ischaemia

- Success over 90%.
- **Complications:** obstruction (stent first), VUR (vesicoureteral reflux; observe or inject), persistent dilatation.

Diagram to draw: dilated ureter above narrow adynamic segment; tapered reimplant.

Key points to remember:

- Megaureter ≥7 mm; POM = adynamic juxtavesical segment.
- Surgery: UTIs, progression, DRF <40% or fall ≥5%.
- Excise, taper, reimplant 5:1; success >90%.

Rapid Revision Sheet – Practice Set 2 Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Iatrogenic ureteric injury (LAQ)

- Distal ureter commonest iatrogenic site; most injuries present late.
- CT with 10-minute delays; retrograde pyelogram is most sensitive and allows stenting.
- Late: stent first; early repair ≤7 days, otherwise delayed after 6 weeks.
- Lower ureter: psoas hitch 95–100%; Boari flap up to 12 cm.

Q2. Discuss mgt. of b/l VUR. (LAQ)

- Bilateral grade IV persists most; bilateral VUR is high risk.
- RIVUR: CAP halves recurrent UTI, not scarring.
- Open reimplant 95.9%; Cohen over 98% and suits bilateral VUR.
- Bilateral extravesical repair: retention up to 10%.

Q3a. Uroflowmetry (SAQ)

- Voided volume ≥150 mL; Qmax is the key value.
- Plateau = fixed obstruction; staccato = sphincter; interrupted = weak detrusor.
- Cannot distinguish BOO from DUA.

Q3b. Ejacultory duct obstruction (SAQ)

- Low volume (<1 mL), acidic, fructose-negative azoospermia with palpable vasa.
- TRUS: vesicle >1.5 cm; aspiration ≥3 sperm per high-power field confirms.
- TURED: sperm return in about half; reflux and epididymitis are its price.

Q3c. Ureterocele (SAQ)

- Upper pole of duplex; never wholly urethral; Chwalla membrane.
- Intravesical vs ectopic predicts TUI success.
- TUI relieves obstruction in 78–97%; new reflux 0–50%.

Q3d. Discuss treatment of b/l VUR. Describe surgical techniques and criteria for their selection (SAQ)

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- Endoscopic Dx/HA (dextranomer/hyaluronic acid) by HIT (hydrodistension implantation): least invasive; lower success at higher grades.
- Bilateral VUR (vesicoureteral reflux): Cohen cross-trigonal reimplant (over 98%) does both sides through one cystotomy; bilateral extravesical (Lich–Gregoir) repair risks retention up to 10%.

Q3e. Darifenacin vs. solifenacin vs. tolterodine (SAQ)

- Darifenacin most M3-selective; tolterodine bladder-selective; solifenacin longest half-life.
- Similar efficacy; STAR: solifenacin beat tolterodine ER.
- Elderly: darifenacin, trospium or β3 agonist; dementia risk 10–65% with >3 months' use.

Q3f. Primary obst. megaureter. (SAQ)

- Megaureter ≥7 mm; POM = adynamic juxtavesical segment.
- Surgery: UTIs, progression, DRF <40% or fall ≥5%.
- Excise, taper, reimplant 5:1; success >90%.